INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD
Department of Artificial Intelligence
AI3403 – MULTI-AGENT SYSTEMS

Self-Healing Distributed Multi-Agent Drone Swarm for Disaster Response

Resilient Task Allocation and Coordination under Dynamic Agent and Communication Failures

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1. Abstract

Disaster-response environments are dynamic and vulnerable to failures in both autonomous agents and communication links. We propose a **self-healing distributed multi-agent system** in which a swarm of drones collaboratively searches a disaster region, detects survivors, shares local information, and dynamically reallocates tasks *without* a centralized controller. The swarm is modeled as a dynamic communication graph; distributed consensus lets agents agree using only neighbor information, while distributed optimization (ADMM) drives task allocation based on distance, battery, workload, and connectivity. On drone or link failure, remaining agents detect the change, update the topology, redistribute unfinished tasks, and continue the mission. The system will be evaluated in simulation using mission, energy, communication, consensus, scalability, and failure-recovery metrics.

2. Motivation and Problem Statement

Centralized disaster-response systems risk a single point of failure: if the controller or a key link goes down, otherwise-functional drones lose coordination. A distributed system instead decides using local observations and neighbor-exchanged information. Our central question:

> *Can a drone swarm maintain coordinated disaster-response operations and autonomously reorganize when drones or communication links dynamically fail?*

The focus is thus on **distributed coordination under dynamic failures** – enabling continued mission progress without any central recovery controller.

3. System Model and Methodology

We simulate a 100×100 disaster region with $\sim$20 drones, 8–10 survivors, obstacles, heterogeneous battery levels, and a dynamically changing communication graph $G(t) = (V, E(t))$. Each drone maintains its position, velocity, battery, current task, neighbor set, local map, and detected-survivor information.

Pipeline: Environment $\rightarrow$ distributed search & survivor detection $\rightarrow$ neighbor information sharing over $G(t) \rightarrow$ {consensus, distributed optimization} $\rightarrow$ coordinated mission execution $\rightarrow$ on failure: detect $\rightarrow$ redistribute tasks $\rightarrow$ reorganize topology $\rightarrow$ resume.

Distributed Consensus: local estimates x_i update as

$$x_i(k+1) = x_i(k) + \alpha \sum_{j \in N_i(k)} a_{ij}\big(x_j(k) - x_i(k)\big),$$

letting the swarm agree on quantities such as search coverage and shared mission estimates.

Distributed Optimization / ADMM: when survivors are found or regions become unassigned, drones jointly pick responders, optimizing distance, battery, workload, connectivity, and region priority – each agent solving a local problem while exchanging coordination variables with neighbors, avoiding a central optimizer.

Failure Handling and Self-Healing: drone failures are detected via lost heartbeats; link failures remove graph edges. Remaining agents update their local view and redistribute unfinished work, restoring mission functionality autonomously.

Resilient Coordination (extension): inconsistent information from faulty agents will be tested against resilient-consensus mechanisms; minimum inter-drone separation and response-time constraints will be enforced, with STL-based verification explored if time permits.

4. Simulation and Evaluation

The system will be implemented as a Python simulation modeling drone motion, survivors, obstacles, battery depletion, communication range, changing topology, and stochastic failures, with visualization of trajectories, links, tasks, and failed agents.

Experiments: (1) centralized vs. distributed allocation; (2) agent-failure robustness at 10/20/30%; (3) communication robustness under degraded connectivity; (4) self-healing recovery time; (5) scalability at 10/20/50(/100) drones; (6) resilient vs. ordinary consensus under inconsistent information.

Metric	Purpose
Mission completion time	Response efficiency
Search coverage / detection rate	Search effectiveness
Task allocation efficiency	Workload distribution
Energy consumption	Battery efficiency
Communication overhead	Coordination cost
Consensus error	Agreement / convergence
Failure recovery time	Self-healing capability
Scalability, Safety violations	Swarm size effect, constraint satisfaction

5. Novelty and Expected Contribution

The novelty lies not in the use of drones but in integrating **distributed coordination, dynamic task allocation, self-healing, and communication resilience** in one framework – combining dynamic communication graphs, consensus, ADMM-based optimization, task redistribution, and autonomous reorganization. The project will provide quantitative comparisons of centralized vs. distributed coordination under normal and failure-prone conditions, characterizing trade-offs among completion time, coverage, energy, communication overhead, consensus convergence, scalability, and recovery.

6. Expected Outcome

A distributed drone swarm that searches a disaster environment, detects survivors, shares information, and allocates tasks dynamically – and, on drone or link failure, detects the change, redistributes tasks, and reorganizes its communication structure without any centralized recovery controller. Core modules: simulation, communication, task allocation, consensus, self-healing. Advanced extensions (progressively integrated): ADMM, resilient consensus, STL-based verification.

7. Team Responsibilities and Timeline

Member	Primary Role	Components
Puli Dinesh	Distributed Optimization	ADMM, formulation, task allocation
Sai Dhanush	Consensus	Consensus, resilient consensus, convergence
Ajay	Failure and Self-Healing	Failure detection, redistribution, recovery
Nenavath Kalyan	Communication	Dynamic graph, links, coordination
Kalyan	Simulation and Evaluation	Environment, movement, visualization, experiments

Timeline:

- **Week 1:** Environment and agent setup – build the simulation world, drone agent model, obstacles, and survivor placement.
- **Week 2:** Communication graph – implement dynamic graph $G(t)$, neighbor discovery, and link range/failure model.
- **Week 3:** Task allocation – distributed assignment of search regions and detected survivors to drones.
- **Week 4:** Consensus – implement and test the distributed consensus update rule across neighbors.
- **Week 5:** Failure detection and self-healing – heartbeat monitoring, edge removal, task redistribution logic.
- **Week 6:** ADMM and resilient consensus – distributed optimization formulation and robustness to faulty agents.
- **Week 7:** Safety and experiments – separation/response-time constraints, run core experiment set.
- **Week 8:** Integration, evaluation, report writing, and final presentation.

A swarm that does not stop when agents or communication links fail — it reorganizes and continues the mission.